PLANETARY STABILITY AND HABITABILITY IN TRIPLE STAR SYSTEMS

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OVERVIEW

Exoplanets
- Definition
- Detection
- Population

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Multiple Star Systems

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Stability

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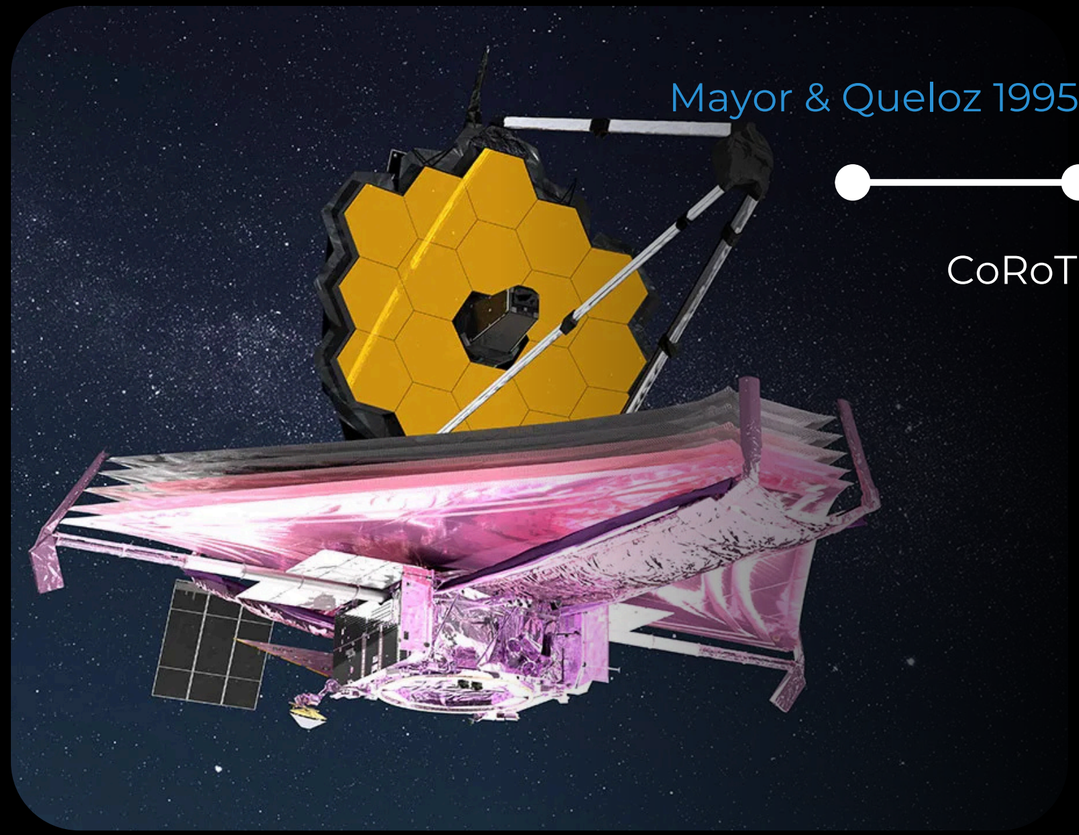
Habitability

Sucerquia Mario



NASA

Mayor & Queloz 1995

Kepler, 2009

TESS, 2018

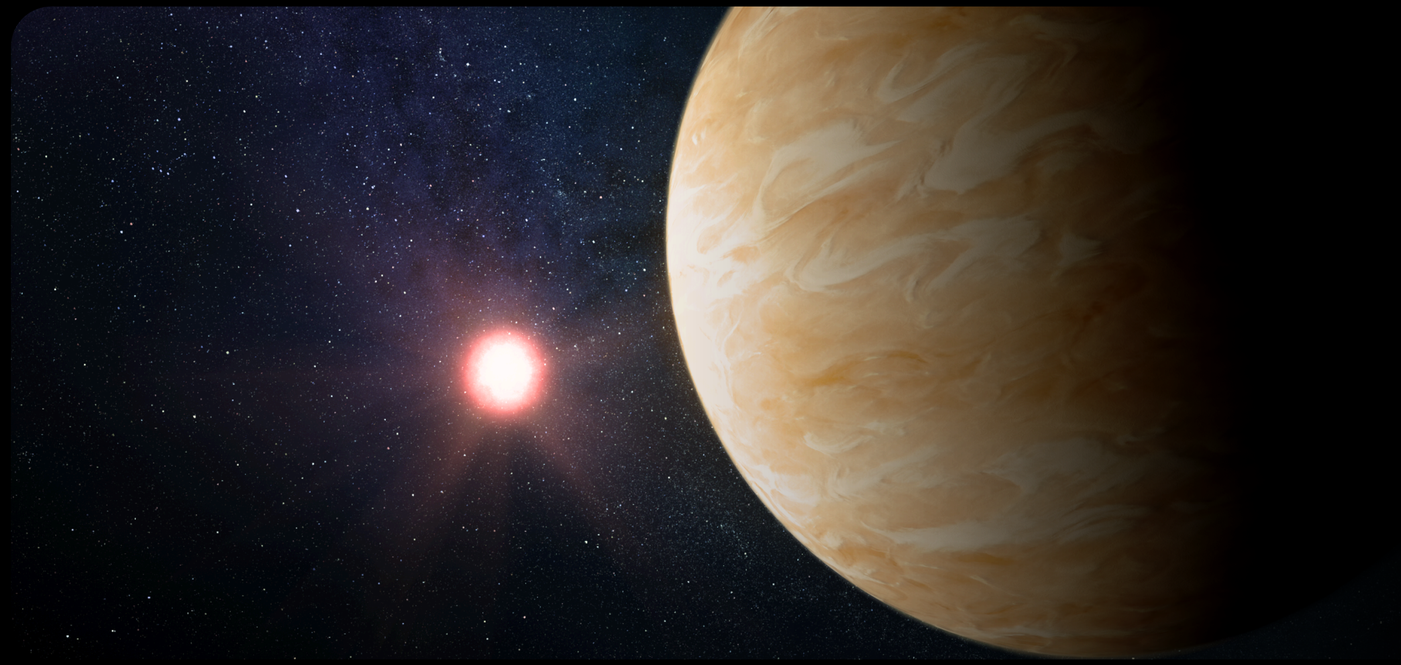
CoRoT, 2006

Gaïa, 2013

JWST, 2021

INTRODUCTION

- ➜ Most stars reside in multiple-star systems (Raghavan et al. 2010)
- ➜ Lack of data for exoplanet in multiple star system
- ➜ Strong need for theoretical and numerical investigation



GJ 1214b, NASA

EXOPLANET DEFINITION

Planetary body orbiting a star or stellar remnant other than the Sun.

➔ Thermonuclear fusion of deuterium mass criterion at 13 M_{Jup} (Burrows et al. 2001)

➔ Orbital clearing criteria (Margot 2015)

Limitations :

- Free-floating planets
- Large moons
- Objects above the 13 M_{Jup} limit without sustained thermonuclear fusion

EXOPLANET DETECTION

Transit

Periodic dimming of the star occurs when a planet crosses its disk. This method measures the planetary radius, orbital period, and inclination, and yields bulk density when combined with radial velocities. Transit spectroscopy can probe atmospheric composition. It favors large, close-in planets.

Radial Velocity

The planet induces Doppler shifts in the stellar spectrum due to mutual orbital motion. This technique measures the projected planetary mass (Mp sin i) and orbital parameters, and is most sensitive to massive, short-period planets.

Direct Imaging

The planet's reflected or emitted light is directly detected by separating it from the host star. This method enables atmospheric characterization and is best suited for young, massive planets on wide orbits.

Astrometry

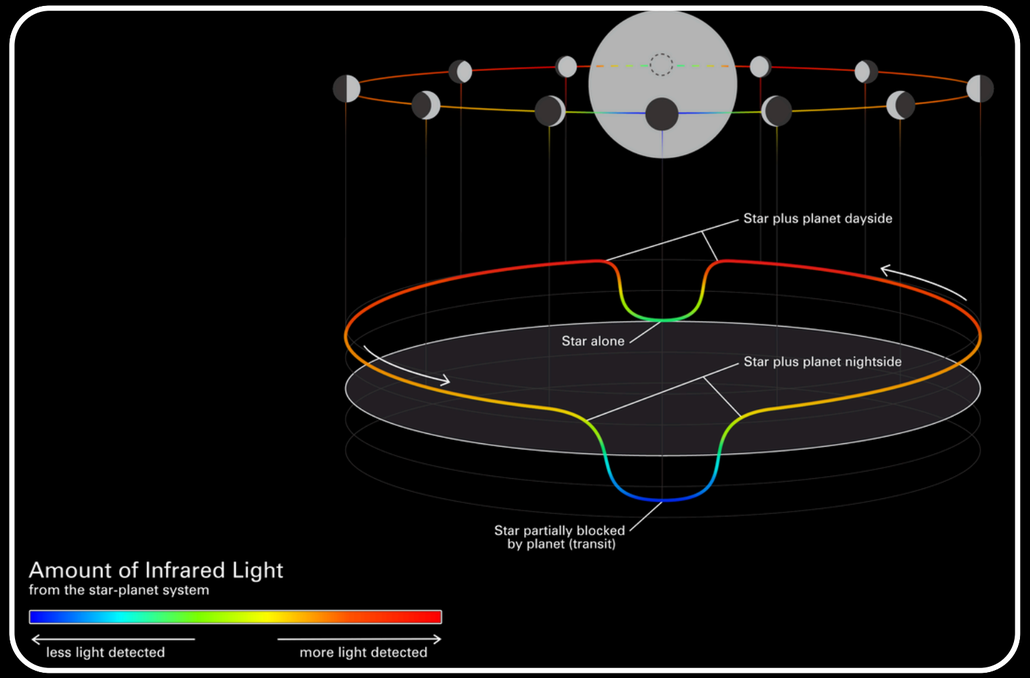
The gravitational pull of the planet causes a small displacement of the star on the sky. Astrometry provides the true planetary mass and full orbital geometry, and is most effective for massive planets on wide orbits.

Microlensing

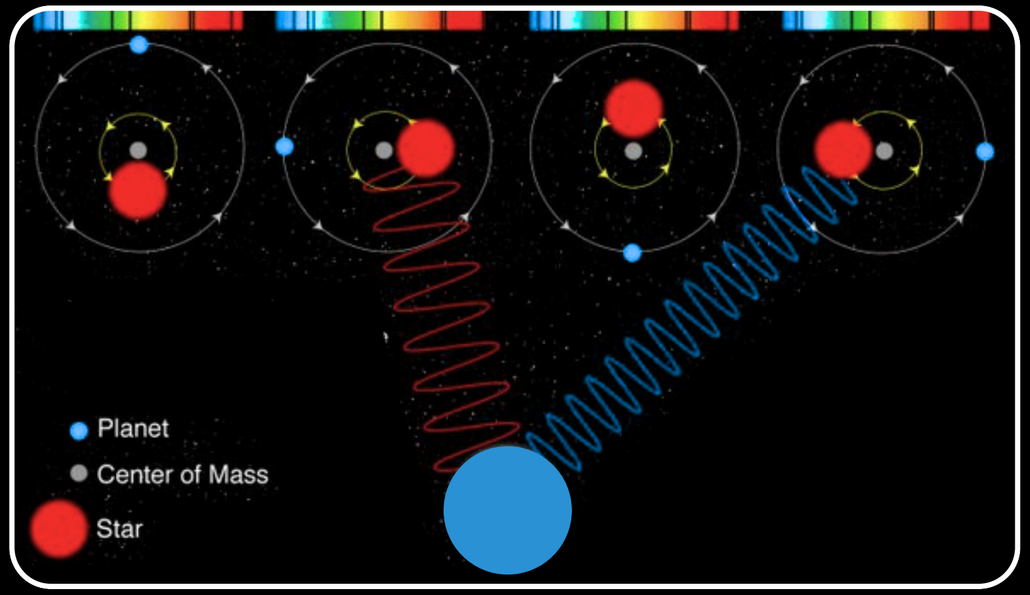
Gravitational lensing by a foreground star produces a short-lived anomaly when a planet is present. Microlensing can detect low-mass and distant planets, including free-floating planets, at separations inaccessible to other methods.

Timing

Variations in periodic signals (e.g. transits or pulses) reveal gravitational perturbations from additional bodies. This method is especially powerful for multi-planet systems, including those in multiple-star environments.



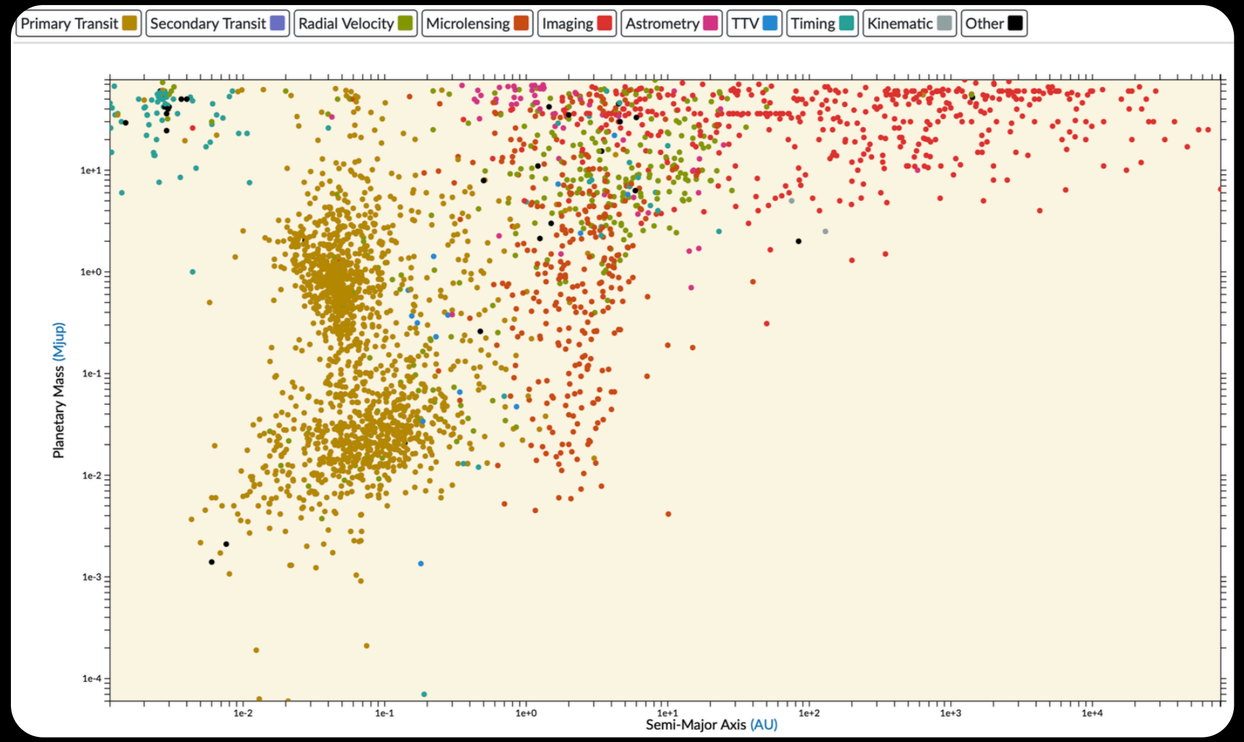
NASA, ESA, CSA, Dani Player (STScI)



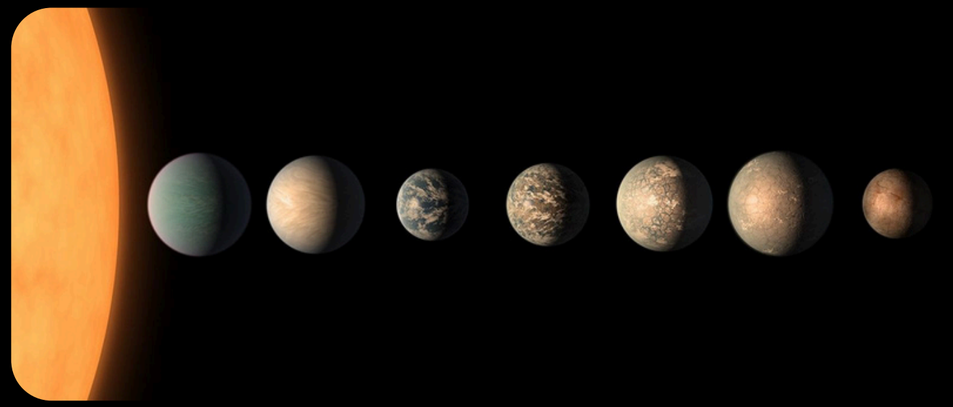
LCOGT.net

Detection method	Number of planets	Fraction (%)
Transit	4716	59.6
Radial velocity	1332	16.8
Direct imaging	1191	15.0
Others	931	11.8
Total	7918	103.2

The Exoplanet Encyclopedia, 02.01.2026



The Exoplanet Encyclopedia, 02.01.2026

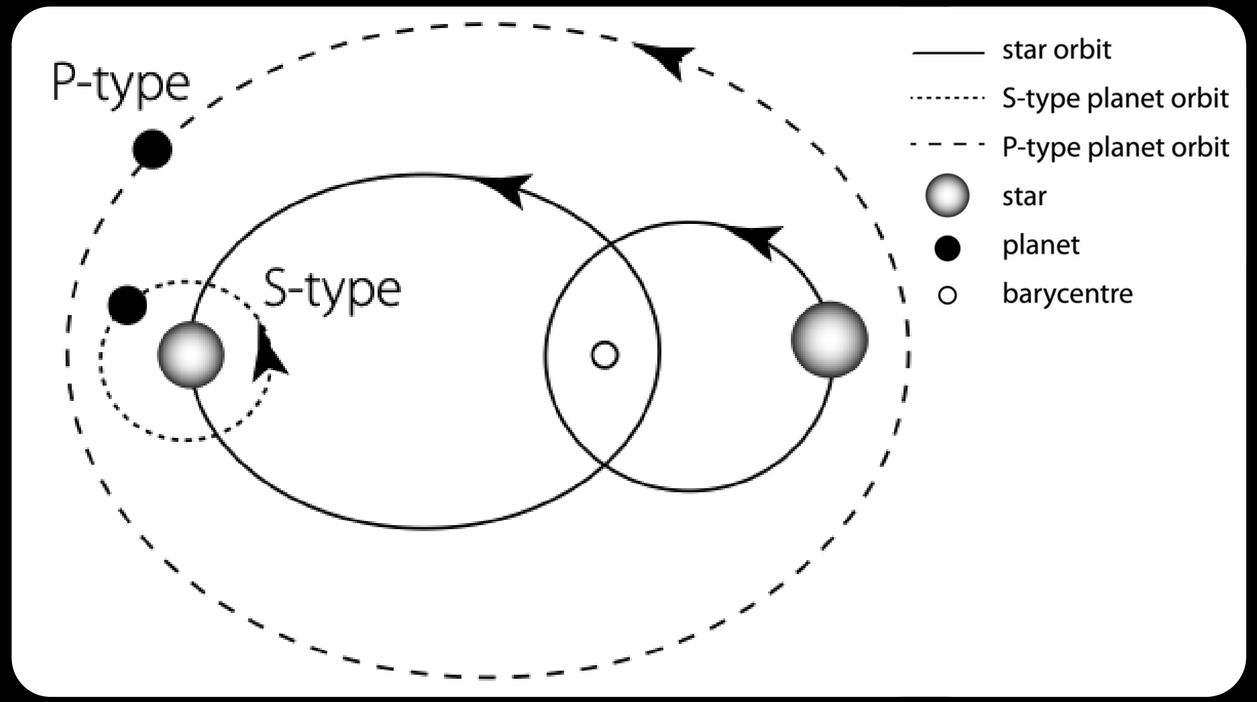


Trappist-1, NASA

- Hot Jupiter
- Cold Jupiter
- Neptune and Super-Earth

EXOPLANET POPULATION

EXOPLANET IN MULTIPLE STAR SYSTEM



Perryman 2018

2 Orbital Configurations

Dvorak 1982

- **Satellite-type :** The planet orbits one of the stars, while the other star acts as a gravitational perturber.
- **Planet-type :** The planet orbits the barycenter of the two stars in the system, such planets are also called circumbinary planets.

824 Confirmed Systems
The Exoplanet Encyclopedia, 02.01.2026

35 Confirmed Systems
The Exoplanet Encyclopedia, 02.01.2026

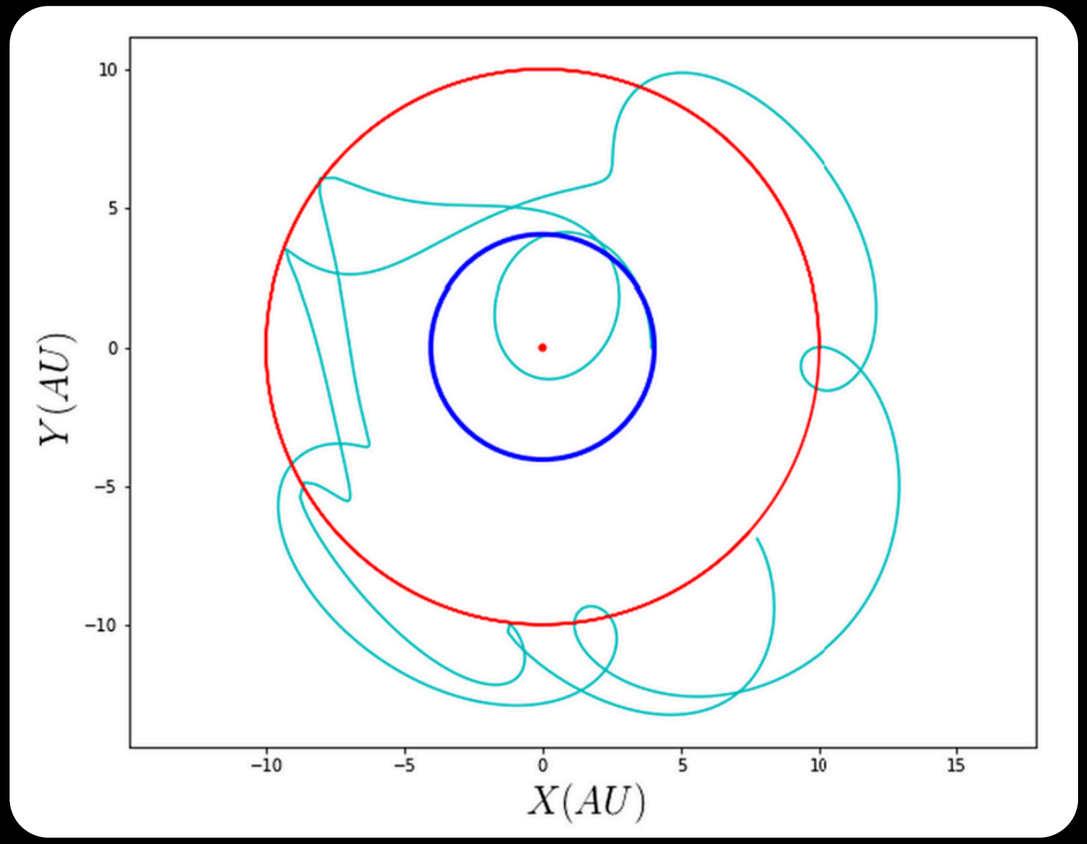
WHY DO ORBITS BECOME UNSTABLE?

N-body (N≥3)→ Intrinsically chaotic

Time-dependant property

Mechanism:

- Resonance overlap (orbital periods → ratio of small integers)
- Cumulative orbital perturbations
- Orbital decay, collision, or ejection

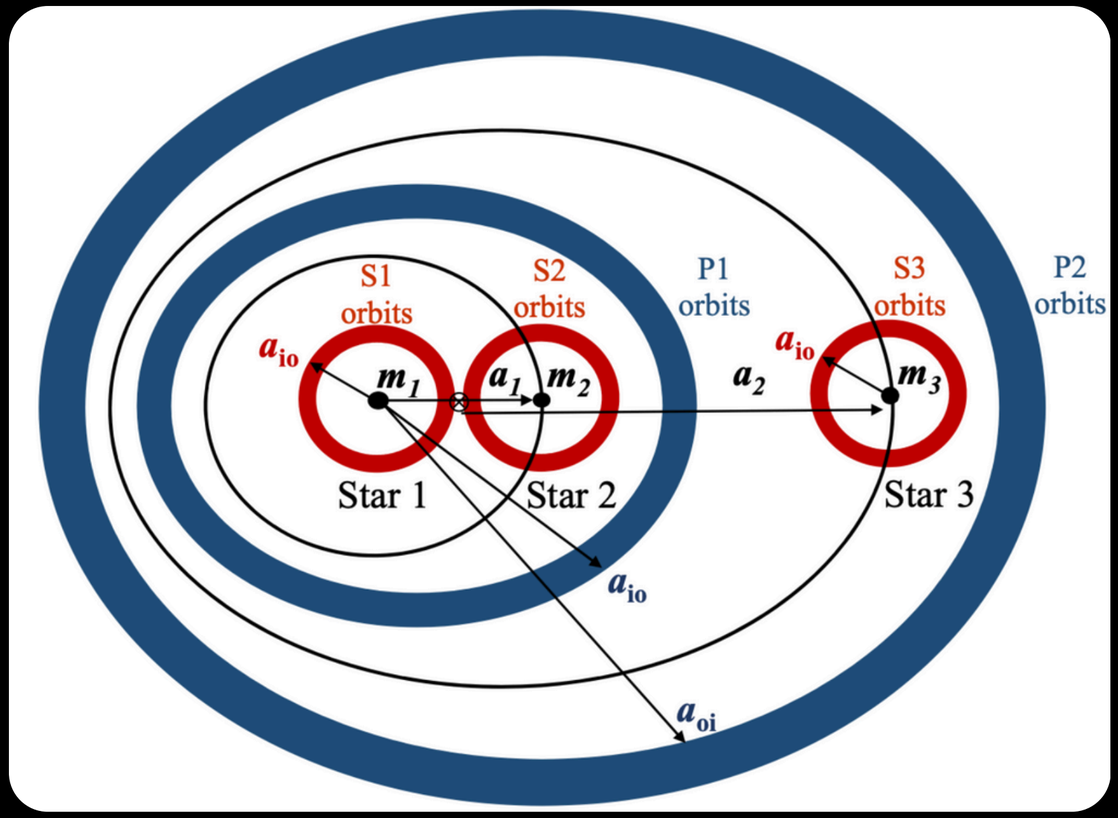


Simulation by De Cesare & Capuzzo-Dolcetta, 2021

Collision of the planet with the secondary star (S-type orbit)

The red circle is the orbit of the secondary star in the frame centered in the primary, the blue circle is the orbit of the unperturbed planetary motion

HIERARCHICAL TRIPLE STAR SYSTEM



Busetti et al. 2018

A hierarchical triple system can be treated as two decoupled binary systems
(Verrier & Evans, 2007)

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- **S-type:** outer band limit, $a<a_{io}$
- **P-type:** inner band limit, $a>a_{oi}$

➔ P1 orbit has an inner and outer limit

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Stability sensitivity:

- Stars' mass ratio and eccentricity of the binary (Holman & Wiegert 1999)

- Inclination
- Retrograde motion (relative to the stars) tend to be more stable than prograde motion (Busetti et al. 2018)

- Planet's mass (De Cesare & Capuzzo-Dolcetta, 2021)

HABITABILITY

Life

A self-sustaining chemical system capable of undergoing Darwinian evolution (NASA 1994).

Habitability

Ability of an environment to support the activity of at least one known organism (Cockell et al. 2016).

Habitable Zone (HZ)

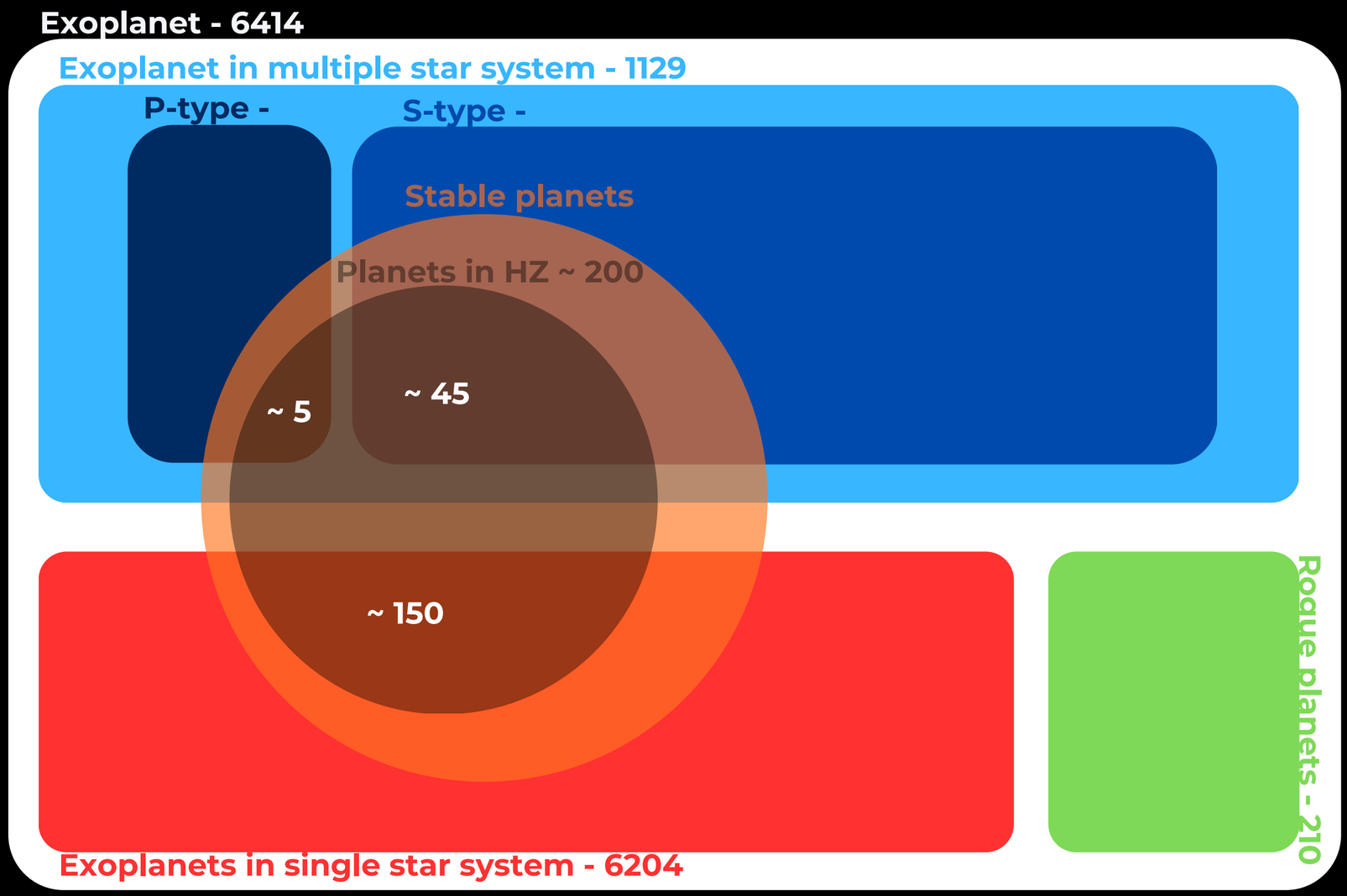
Range of orbital distances around a main-sequence star within which liquid water can exist on the surface of a rocky planet (Kasting et al. 1993). The location and extent of the habitable zone depend strongly on stellar luminosity and spectral energy distribution which depend on star mass.

Extension

- ➜ Improved atmospheric radiative transfer and thermal model (Kopparapu et al. 2013)
- ➜ Albedo effects (Selsis et al. 2007)
- ➜ Temporal evolution of stellar luminosity (Rushby et al. 2013)
- ➜ Dependence on planetary mass (Kopparapu et al. 2014)

Limitations

- Intense X-ray and UV stellar irradiation
- Moons with liquid water beneath icy shells (eg. Europa, Enceladus) (McMahon et al. 2013)
- Rogue planets (Abbot & Switzer 2011)
- Venus-like planets located in HZ (Cockell 1999)
- Biological activity influence (Lovelock & Margulis 1974)



The Exoplanet Encyclopedia, Burrows et al. 2001 mass criteria, 14.01.2026

Habitable → Stable

HZ is **time dependent** :

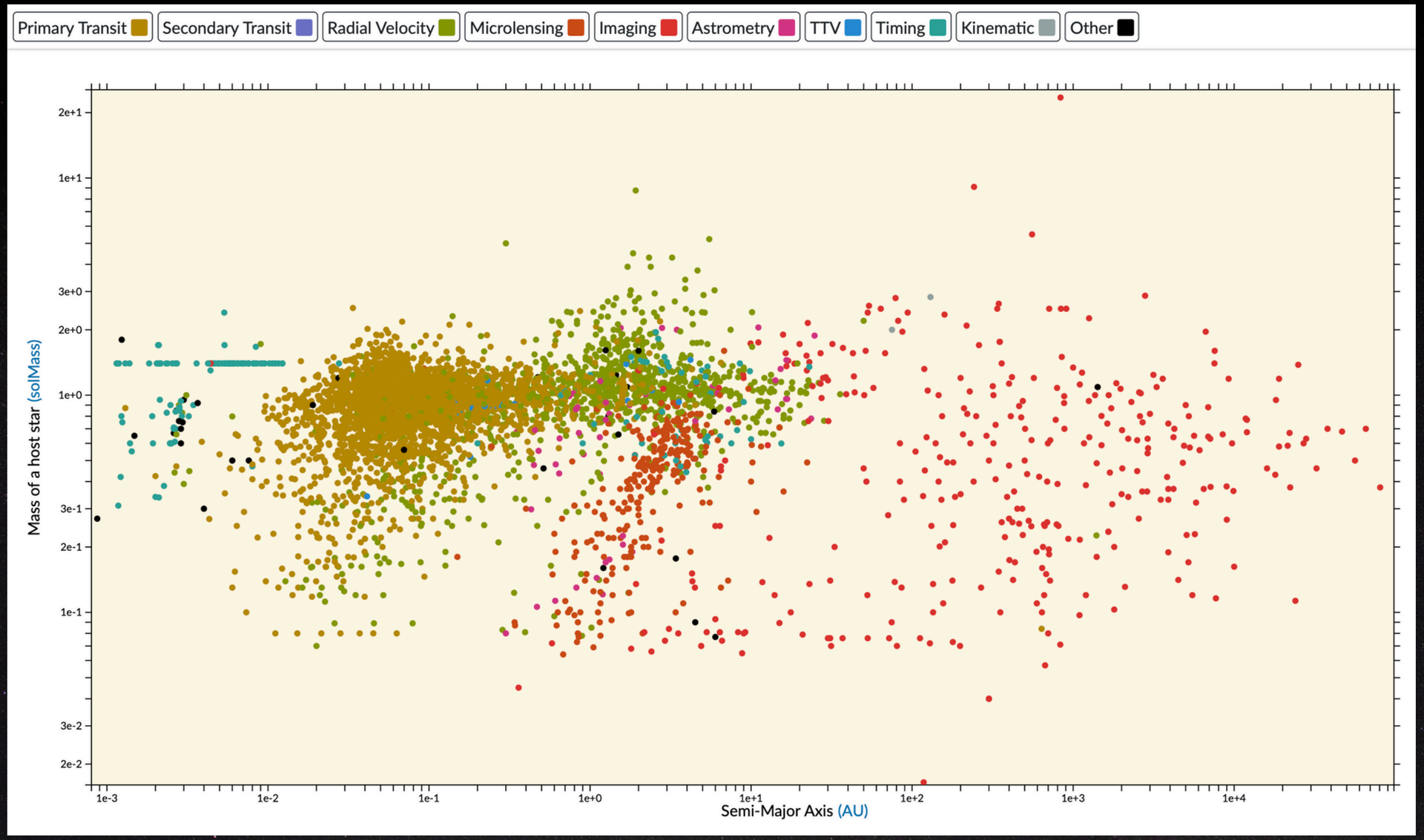
- Permanent Habitable Zone (PHZ)
- Extended Habitable Zone (EHZ)
- Averaged Habitable Zone (AHZ)

Eggl et al. 2012

CONCLUSION – UPCOMING WORK

- ➜ Numerical simulations using REBOUND (Python library)
- ➜ Exploring stability parameters (mass, eccentricity, etc.)
- ➜ Calculating stellar flux
- ➜ Stability-habitability map

THANK YOU!



The Exoplanet Encyclopedia, 14.01.2026

- inner HZ (runaway greenhouse) : 0.95 AU × $(L^*/Lsun)^{0.5}$
- outer HZ (maximum greenhouse) : 1.67 AU × $(L^*/Lsun)^{0.5}$
  Kopparappu 2013
- $L \propto M^{3.5}$ pour les étoiles de type FGK

- a (HZ,inner) = 0.95 $M^{*1.75}$ AU
- a (HZ,outer) = 1.67 $M^{*1.75}$ AU